Functional Requirements Document – My Profile

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Functional Requirements Document (sub-module 07.1 My Profile, code `PROF`) · **Version:** 1.0 · **Status:** Draft **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026 **Author:** Jojo George (jojogeorge3344@gmail.com) **Based on:** BRD-My-Profile.md v1.0 © 2026 Jojo George. All rights reserved.

Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version: My Profile, Change Password and self-service two-step verification, traced to BRD-My-Profile v1.0

Related Documents

Document	Purpose	Location
BRD – My Profile	Business needs this FRD details (`REQ-PROF-nn`, `BR-PROF-nn`)	`../../../BRD/07-My-Profile/07.1-My-Profile/BRD-My-Profile.md`
BRD / FRD – Overall	Roles, global rules `BR-nn`, global NFRs, Authentication (Login, Security Setup, Account Recovery)	`../../../BRD/00-Overall/BRD-TaskFlow-Overall.md`, `../../00-Overall/FRD-TaskFlow-Overall.md`
FRD – Employee Management	Administrator-managed account fields, administrator two-step reset	`../../03-TaskFlow-Management/03.1-Employee-Management/FRD-Employee-Management.md`
FRD – TFA Assistant	Action Card that prepares an own-profile change	`../../08-TFA-Assistant/08.1-Assistant/FRD-Assistant.md`
Development Plan	Screen 19 – User Profile	`TF - Documents/Screens Wise Feature Development Plan/19-user-profile-screen.md`
HLD / LLD / TDD	Technical design	`../../../HLD/07-My-Profile/07.1-My-Profile/HLD-My-Profile.md`, `../../../LLD/07-My-Profile/07.1-My-Profile/LLD-My-Profile.md`, `TF - Documents/TF - TDD/TDD-TaskFlow.md`

1. Introduction

1.1 Purpose

This FRD specifies the detailed, testable behaviour of the My Profile sub-module: the My Profile screen, the Change Password screen and the self-service two-step verification actions used by the Security Setup screen. It turns every

`REQ-PROF-nn` and `BR-PROF-nn` of the My Profile BRD into functional requirements `FR-PROF-nn` with inputs, outputs, validations, messages and permissions.

1.2 Scope

In scope: viewing the own account; editing full name, email address and mobile number; changing the own password with an authenticator or backup code; starting authenticator setup, turning two-step verification on, receiving backup codes; regenerating backup codes and checking a code (server actions without a screen); the profile rules applied to a TFA own-profile change.

Out of scope (BRD §3.2): another user's profile (Employee Management); changing own role, projects, departments, username, employee code or status; profile photo; forgotten-password recovery and the sign-in flow (Authentication, overall FRD); the layout and step order of the Security Setup screen (Authentication, overall FRD); turning two-step verification off by the user.

1.3 Audience

Developer and tester of TaskFlow, reviewers, and the product owner (Jojo George).

1.4 Conventions

- Requirement IDs: `FR-PROF-nn`. Use cases: `UC-PROF-nn`. Business rules: `BR-PROF-nn` (sub-module BRD) and `BR-nn` (overall BRD).
- BRD trace: `REQ-PROF-nn` (sub-module BRD §5).
- Priority: **Must / Should / Could** (MoSCoW).
- "The system shall …" = mandatory behaviour.
- Text in `{braces}` inside backticks is a value filled in at run time.

1.5 Definitions & Abbreviations

Term	Meaning
Own scope	The user acts only on their own account; the account is always taken from the sign-in session
Two-step verification (MFA)	A second factor from an authenticator app (6-digit time-based code) or a single-use backup code
Backup code	One of 10 single-use codes issued when two-step verification is turned on
Setup pending	State of a signed-in user who has not yet turned on two-step verification
Always On	A permission held by every role that cannot be removed in Role Configuration
TFA	TaskFlow Assistant, the built-in AI assistant

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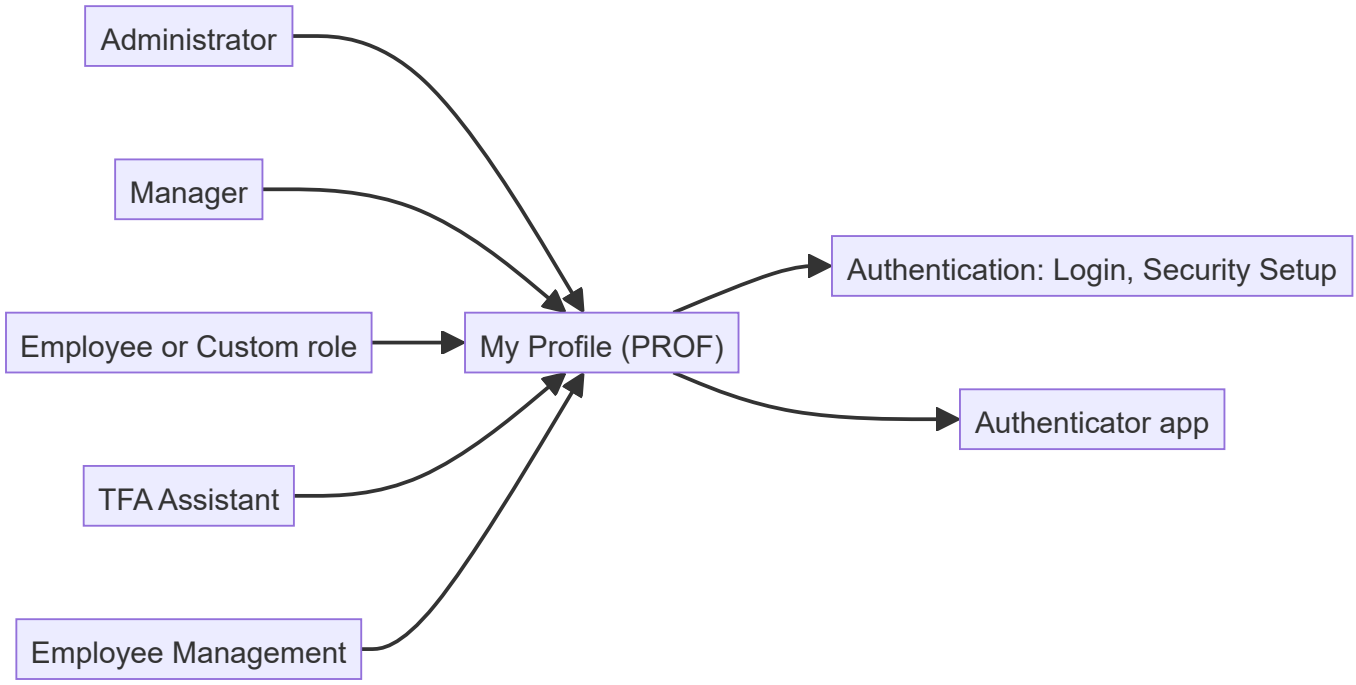
2. System Overview

2.1 Module Map

Code	Module	Child modules / screens	Primary roles
PROF	My Profile (07.1)	My Profile, Change Password, Two-step verification self-service (used by Security Setup)	Administrator, Manager, Employee, Custom role
AUTH	Authentication (overall FRD) – interface only	Login, Security Setup, Account Recovery	All users

Code	Module	Child modules / screens	Primary roles
EMP	Employee Management – interface only	Maintains role, projects, departments, status; administrator two-step reset	Administrator
TFA	TFA Assistant – interface only	Action Card "Update your profile", password guidance	Holders of `assistant.chat`

2.2 Context Diagram



3. User Roles & Permissions

3.1 Roles

Role	Description	Data scope
Administrator	Full system user; maintains own profile, password and authenticator here (other accounts in Employee Management)	Own
Manager	Leads teams and projects; also sees own Managed projects and Team	Own
Employee	Works on assigned tasks	Own
Custom role	Role created in Role Configuration; same profile rights as Employee	Own

3.2 Permission Matrix (functional view)

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
My Profile › View my profile	`profile.view`	Always On	Always On	Always On	Always On

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
My Profile › Edit my personal information	`profile.edit`	Always On	Always On	Always On	Always On
My Profile › Change my password	`profile.password.change`	Always On	Always On	Always On	Always On
My Profile › Manage my authenticator (setup, backup codes)	`profile.mfa.manage`	Always On	Always On	Always On	Always On

The four keys are given to every user automatically, are not stored per role and cannot be assigned or removed in Role Configuration (`BR-PROF-05` , global `BR-05`). The **My Profile** menu item is always shown.

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4. Functional Requirements by Module

4.1 Module: My Profile (PROF)

Purpose: Self-service maintenance of the signed-in user's own account: contact details, password and second factor. · **BRD reference:** BRD-My-Profile §5 SCREEN 1–3, `BR-PROF-01` – `BR-PROF-13` ; overall `BR-03` , `BR-05` , `BR-24` , `BR-25` , `BR-28`

4.1.1 Screen: My Profile

Field	Value
Route / location	`/profile`
Access	Administrator, Manager, Employee, Custom role – `profile.view` , `profile.edit` (Always On); own account only
Entry points	Side menu **My Profile**; name/avatar link in the top bar; TFA link button **My Profile**; reachable while setup is pending

Screen elements

Element	Type	Description	Mandatory	Visible to
Avatar	Initials	Initials of the full name	–	All users
Full name, `@username` , email	Header text	Identity of the signed-in user	–	All users
Summary chips	Chips	Role; Active/Inactive; one chip per project allocation; "Two-step on"/"Two-step off"; "Last sign-in …" when a last sign-in exists	–	All users
Account details	Read-only card	Username, Employee code, Role, Projects with total "% allocated", Departments, Managers, Account status; note "Your role, project and status are managed by an administrator."	–	All users
Managed projects, Team	Read-only rows	Projects the user manages and the user's team	–	Manager

Element	Type	Description	Mandatory	Visible to
Full name	Text input	Own full name	Yes	All users
Email address	Text input + fixed suffix	Part before the fixed `@taskflow.com` suffix	Yes	All users
Mobile number	Text input	International format, e.g. `+919876543210`	No	All users
Save changes	Button	Saves the personal information; shows "Saving…" while saving and "Saved" afterwards	–	All users
Password card – Change password	Button/link	Opens the Change Password screen	–	All users
Loading placeholders	Skeleton	Shown while the account loads	–	All users
Error state – Retry	Message + button	Shown when the account cannot be loaded	–	All users

Functional requirements

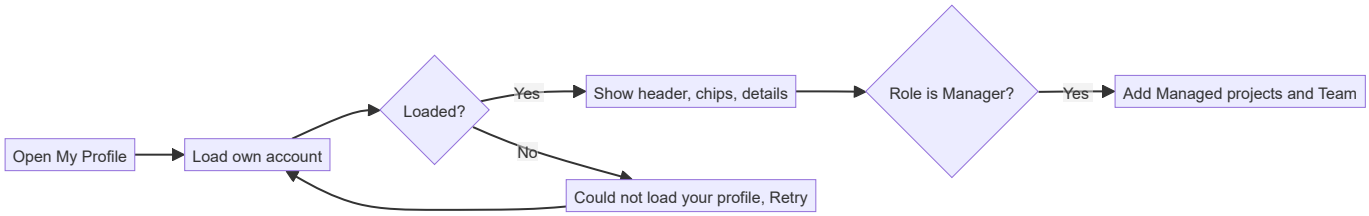
ID	Requirement	Priority	Permission	BRD trace
FR-PROF-01	The system shall load the account of the signed-in user taken from the sign-in session, and shall offer no way to choose or open another user's profile.	Must	`profile.view`	REQ-PROF-01, BR-PROF-01
FR-PROF-02	The system shall show the header (initials avatar, full name, `@username`, email) and the summary chips: role, Active/Inactive, one chip per project allocation, "Two-step on"/"Two-step off", and "Last sign-in …" when a last sign-in time exists.	Must	`profile.view`	REQ-PROF-02
FR-PROF-03	The system shall show Username, Employee code, Role, Projects with the total "% allocated", Departments, Managers and Account status as read-only values with the note "Your role, project and status are managed by an administrator."	Must	`profile.view`	REQ-PROF-02, BR-PROF-02
FR-PROF-04	The system shall additionally show the rows Managed projects and Team when the user's role is Manager, and shall not show them for other roles.	Should	`profile.view`	REQ-PROF-03
FR-PROF-05	The system shall offer exactly three editable fields – Full name, Email address and Mobile number – pre-filled with the current values, and shall ignore any attempt to change role, projects, departments, username, employee code or status through the profile save.	Must	`profile.edit`	REQ-PROF-04, BR-PROF-02
FR-PROF-06	The system shall validate each editable field while the user types and show the message next to the field (rules in "Field validations").	Must	`profile.edit`	REQ-PROF-05

ID	Requirement	Priority	Permission	BRD trace
FR-PROF-07	The system shall validate all fields again when **Save changes** is clicked and save nothing when any rule fails, showing the error above the form and next to the field concerned.	Must	`profile.edit`	REQ-PROF-05
FR-PROF-08	The system shall build the email from the typed part plus the fixed suffix `@taskflow.com`, trim it and store it in lower case.	Must	`profile.edit`	REQ-PROF-06, BR-PROF-03, BR-24
FR-PROF-09	The system shall refuse an email whose part before `@` matches that of another account, including deleted accounts, with "Email already exists."	Must	`profile.edit`	REQ-PROF-07, BR-PROF-03
FR-PROF-10	The system shall refuse a save when another user has both the same email and the same mobile number, with "A user with this email and mobile number already exists.", and shall accept a mobile number shared with colleagues otherwise.	Must	`profile.edit`	REQ-PROF-08, BR-PROF-04
FR-PROF-11	The system shall remove spaces, dashes, dots and brackets from the mobile number before checking it, and shall remove the saved mobile number when the field is left empty.	Should	`profile.edit`	REQ-PROF-09
FR-PROF-12	The system shall, after a successful save, show the toast "Profile updated", show "Saved" next to the button and update the user's name in the top bar without a page reload.	Must	`profile.edit`	REQ-PROF-10
FR-PROF-13	The system shall refuse a save when the record was changed by someone else since it was loaded, with "Someone else changed this record while you were saving. Reload it and try again."	Must	`profile.edit`	REQ-PROF-05, NFR-PROF-05
FR-PROF-14	The system shall show "Could not load your profile." with a **Retry** button when the account cannot be loaded, and Retry shall load it again.	Should	`profile.view`	REQ-PROF-11
FR-PROF-15	The system shall keep My Profile viewable and saveable while the user's two-step setup is pending; Security Setup step 1 (Your details) shall save through the same rules.	Must	`profile.view`, `profile.edit`	REQ-PROF-12, BR-PROF-05, BR-03
FR-PROF-16	The system shall apply FR-PROF-07 to FR-PROF-10 and FR-PROF-13 when a TFA "Update your profile" card is confirmed, change only the name and/or email named on the card, and leave the mobile number unchanged.	Should	`profile.edit` (+ `assistant.actions.confirm` on the card)	REQ-PROF-13, BR-PROF-13
FR-PROF-17	The system shall create no notification, mail or task activity entry for a profile save.	Should	`profile.edit`	BR-PROF-12
FR-PROF-18	The system shall open the Change Password screen when **Change password** on the Password card is clicked.	Must	`profile.password.change`	SCREEN 1 elements (Password card), REQ-PROF-15

Use case: UC-PROF-01 – View my profile

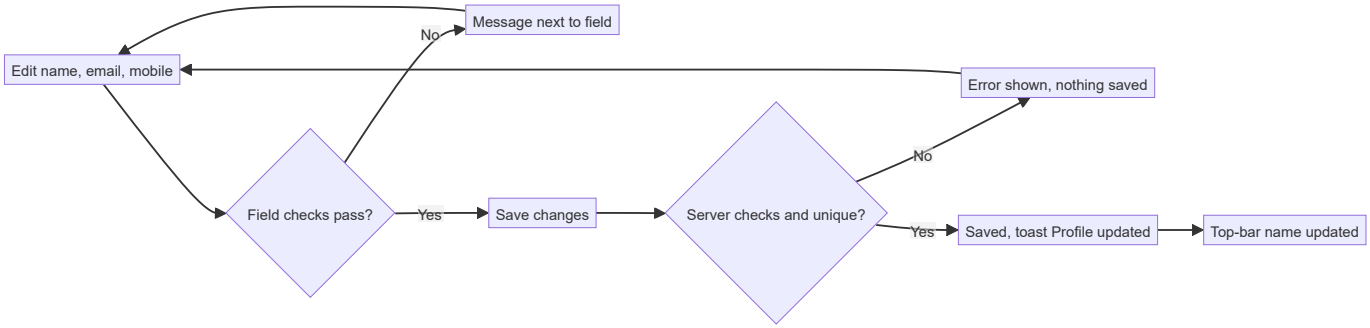
Item	Detail
Actor	Any signed-in user (Administrator, Manager, Employee, Custom role)
Preconditions	Signed in; `profile.view` (Always On)

Item	Detail
Trigger	Clicks **My Profile** in the side menu or the name/avatar in the top bar
Main flow	1. System shows loading placeholders. 2. System loads the own account from the session. 3. System shows header, chips and Account details. 4. For a Manager, the system also shows Managed projects and Team.
Alternate flows	A1: user opens the screen while setup is pending – the screen opens normally (FR-PROF-15).
Exceptions	E1: account cannot be loaded → "Could not load your profile." with **Retry** (FR-PROF-14).
Postconditions	Own account shown; nothing changed



Use case: UC-PROF-02 – Edit personal information

Item	Detail
Actor	Any signed-in user
Preconditions	My Profile loaded; `profile.edit` (Always On)
Trigger	Changes Full name, Email address and/or Mobile number and clicks **Save changes**
Main flow	1. System checks each field while typing. 2. User clicks **Save changes**; button shows "Saving…". 3. System re-checks format, length and uniqueness. 4. System stores the email in lower case on `@taskflow.com` and the normalised mobile number. 5. System shows "Profile updated" and "Saved" and updates the top-bar name.
Alternate flows	A1: Mobile number cleared → saved number removed (FR-PROF-11). A2: Started from Security Setup step 1 → same rules, then setup continues (FR-PROF-15).
Exceptions	E1: invalid field → field message, nothing saved. E2: email used by another or a deleted account → "Email already exists.". E3: same email and mobile as another user → "A user with this email and mobile number already exists.". E4: concurrent change → "Someone else changed this record while you were saving. Reload it and try again.". E5: other failure → "Could not update your profile." / "Something went wrong. Please try again."
Postconditions	Contact details saved; no notification, mail or activity entry (FR-PROF-17)



Field validations

Field	Rule	Error message
Full name	Required; spaces at start and end trimmed; max 200 characters	"Full name is required." / "This field is too long."
Email address	Required	"Email is required."
Email address (screen)	Part before `@taskflow.com`: letters, digits, dot, underscore or hyphen; no leading, trailing or double dots; max 64 characters	"Use letters, digits, dot, underscore or hyphen before @taskflow.com."
Email address (server)	Full address max 256 characters, fixed domain `@taskflow.com`	"Email must be a name using letters, digits, dot, underscore or hyphen followed by @taskflow.com."
Email address	Unique by the part before `@`, deleted accounts included	"Email already exists."
Mobile number (screen)	Optional; international format	"Use international format, for example +919876543210."
Mobile number (server)	After removing spaces, dashes, dots and brackets: plus sign, 8–15 digits, no leading zero after the plus	"Mobile number must be in international format, for example +919876543210."
Email + Mobile number	Pair must not match another user	"A user with this email and mobile number already exists."

Business rules applied: BR-PROF-01, BR-PROF-02, BR-PROF-03, BR-PROF-04, BR-PROF-05, BR-PROF-12, BR-PROF-13; overall BR-03, BR-05, BR-24.

Notifications / side effects

Event	Who is notified	Channel
Profile saved	Nobody (BR-PROF-12)	– (top-bar name refreshed for the user only)

Messages

Situation	Message
Success	"Profile updated" (toast) · "Saved" (next to the button)
Save failed, no specific reason	"Could not update your profile."
Load failed	"Could not load your profile."
Concurrent change	"Someone else changed this record while you were saving. Reload it and try again."
Unexpected error	"Something went wrong. Please try again."
No permission	*Not applicable – all profile permissions are Always On.*

4.1.2 Screen: Change Password

Field	Value
Route / location	`/profile/change-password`
Access	Administrator, Manager, Employee, Custom role – `profile.password.change` (Always On); own account; two-step verification must be on

Field	Value
Entry points	My Profile → Password card → **Change password**; TFA password guidance links to My Profile

Screen elements

Element	Type	Description	Mandatory	Visible to
Back to profile	Link	Returns to My Profile	–	All users
New password	Password input + show/hide	New password with strength meter Weak / Fair / Good / Strong	Yes	All users
Confirm new password	Password input + show/hide	Repeats the new password	Yes	All users
Hint	Text	"At least 6 and at most 128 characters. Mix upper and lower case, numbers and symbols for a stronger one."	–	All users
Change password	Button	Checks the form and opens the code popup	–	All users
"Confirm it is you" popup – Authenticator code	Text input	6-digit authenticator code or an unused backup code	Yes	All users
Cancel	Button	Closes the popup	–	All users
Confirm	Button	Sends the change; shows "Changing…" while working	–	All users

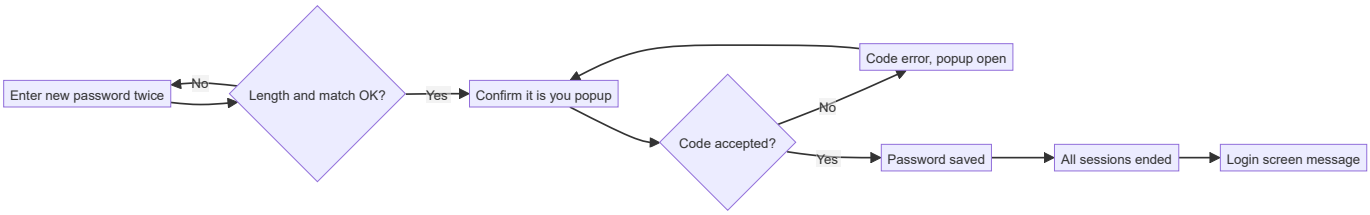
Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-PROF-19	The system shall accept a new password of 6 to 128 characters on the screen and on the server, the same range as sign-in.	Must	`profile.password.change`	REQ-PROF-14
FR-PROF-20	The system shall require both password fields, require them to match, and open the code popup only when the form is valid, sending nothing before that.	Must	`profile.password.change`	REQ-PROF-15
FR-PROF-21	The system shall update the strength meter (Weak, Fair, Good, Strong) while the user types, as guidance only; only the length rule shall be enforced.	Could	`profile.password.change`	REQ-PROF-16
FR-PROF-22	The system shall confirm the change with a 6-digit authenticator code or an unused backup code entered in the "Confirm it is you" popup, and shall not ask for the current password.	Must	`profile.password.change`	REQ-PROF-17, BR-PROF-06
FR-PROF-23	The system shall refuse the change while two-step verification is not on, with "Set up two-step verification before changing your password."	Must	`profile.password.change`	REQ-PROF-18, BR-PROF-07
FR-PROF-24	The system shall refuse an authenticator or backup code that has already been accepted once.	Must	`profile.password.change`	REQ-PROF-19, BR-PROF-10

ID	Requirement	Priority	Permission	BRD trace
FR-PROF-25	The system shall lock code checks for 5 minutes after 5 wrong codes in a row, using the same counter as sign-in, with "Too many wrong codes. Try again in a few minutes."	Must	`profile.password.change`	REQ-PROF-20, BR-PROF-09
FR-PROF-26	The system shall, after a successful change, end every session of the user on every device including the current browser, and open the Login screen with "Your username or password was changed. Sign in again with your new details."	Must	`profile.password.change`	REQ-PROF-21, BR-PROF-08, BR-25
FR-PROF-27	The system shall keep the popup open with the code error when the code is wrong or expired, and close the popup and show the error on the form when the password itself is rejected.	Should	`profile.password.change`	REQ-PROF-22
FR-PROF-28	The system shall close the popup without changing the password when **Cancel** is clicked, and return to My Profile when **Back to profile** is clicked.	Should	`profile.password.change`	REQ-PROF-15, REQ-PROF-22
FR-PROF-29	The system shall send no notification or mail for a password change.	Should	`profile.password.change`	BR-PROF-12

Use case: UC-PROF-03 – Change my password

Item	Detail
Actor	Any signed-in user with two-step verification on
Preconditions	Signed in; `profile.password.change` (Always On); two-step verification on
Trigger	Clicks **Change password** on the Password card
Main flow	1. User enters New password and Confirm new password. 2. User clicks **Change password**; system checks length and match. 3. System opens "Confirm it is you". 4. User enters a 6-digit or backup code and clicks **Confirm** ("Changing…"). 5. System checks two-step is on and the code is valid and unused. 6. System saves the new password. 7. System ends every session and opens Login with "Your username or password was changed. Sign in again with your new details."
Alternate flows	A1: user enters a backup code instead of an app code – accepted once (FR-PROF-24). A2: **Cancel** – popup closes, nothing changed.
Exceptions	E1: length or mismatch → field message, popup not opened. E2: code wrong or expired → "The code is incorrect or has expired.", popup stays open. E3: 5 wrong codes → "Too many wrong codes. Try again in a few minutes.". E4: two-step not on → "Set up two-step verification before changing your password.". E5: password rejected → error on the form ("Could not change your password." when no reason).
Postconditions	New password stored as a hash; all sessions ended; user signs in again



Field validations

Field	Rule	Error message
New password, Confirm new password	Required	"Required."
New password	At least 6 characters	"Password must be at least 6 characters."
New password	At most 128 characters	"Password must be at most 128 characters."
Confirm new password	Equal to New password	"Passwords do not match."
Authenticator code	Required	"Enter the code from your authenticator app."
Authenticator code	Valid, unexpired, not used before	"The code is incorrect or has expired."

Business rules applied: BR-PROF-01, BR-PROF-06, BR-PROF-07, BR-PROF-08, BR-PROF-09, BR-PROF-10, BR-PROF-12; overall BR-25, BR-28.

Notifications / side effects

Event	Who is notified	Channel
Password changed	Nobody; the user's sessions on every device end	–

Messages

Situation	Message
Success	Login screen: "Your username or password was changed. Sign in again with your new details."
Too many wrong codes	"Too many wrong codes. Try again in a few minutes."
Two-step not on	"Set up two-step verification before changing your password."
Code rejected, no reason	"That code did not work. Try the new code from your app."
Password rejected, no reason	"Could not change your password."
No permission	*Not applicable – Always On; while setup is pending the screen is blocked with "Set up two-step verification to continue." (FR-PROF-36).*

4.1.3 Screen: Two-Step Verification (self-service actions used by Security Setup)

Field	Value
Route / location	`/security-setup` (screen owned by Authentication, see overall FRD); status chip on `/profile`
Access	Administrator, Manager, Employee, Custom role – `profile.mfa.manage` (Always On); own account
Entry points	Opened automatically after sign-in while setup is pending; every blocked screen redirects there

Screen elements (only the elements that call the account actions of this sub-module)

Element	Type	Description	Mandatory	Visible to
Current password	Password input + Show/Hide	Starts setup	Yes	Users with setup pending
Show my QR code	Button	Starts setup after the password check	–	Users with setup pending
QR code and setup key, Copy key	Image, text, button	New authenticator key	–	Users after setup start
6-digit code	Text input	First code from the new authenticator	Yes	Users after setup start
Verify and turn on	Button	Turns two-step verification on	–	Users after setup start
Backup codes list – Copy all, Download	List, buttons	10 backup codes shown once; download file `taskflow-backup-codes.txt`	–	Users who just turned two-step on
"Two-step on"/"Two-step off" chip	Chip on My Profile	Current status	–	All users

Functional requirements

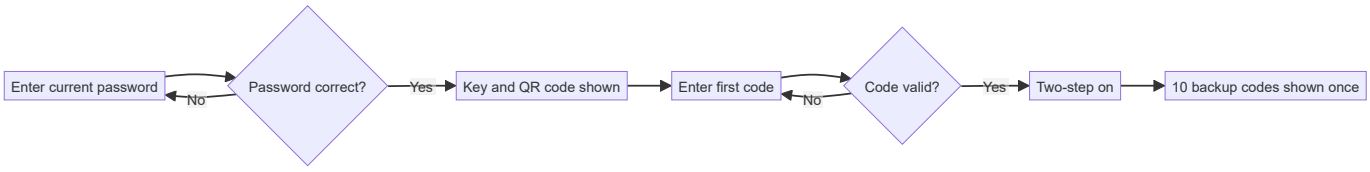
ID	Requirement	Priority	Permission	BRD trace
FR-PROF-30	The system shall start authenticator setup only after the user's current password is checked, allowing 5 password attempts in 5 minutes per account.	Must	`profile.mfa.manage`	REQ-PROF-23
FR-PROF-31	The system shall, when setup starts, show a new key and QR code for a standard authenticator app while two-step verification stays off.	Must	`profile.mfa.manage`	REQ-PROF-23
FR-PROF-32	The system shall refuse to start setup while two-step verification is already on, with "Authenticator MFA is already on. Turn it off first to set it up again."	Must	`profile.mfa.manage`	REQ-PROF-24
FR-PROF-33	The system shall turn two-step verification on only after the first valid code from the new authenticator, and refuse a code sent before setup was started with "Start the setup first to get a key."	Must	`profile.mfa.manage`	REQ-PROF-25
FR-PROF-34	The system shall issue 10 single-use backup codes when two-step verification is turned on, show them only once, and offer Copy all and Download (`taskflow-backup-codes.txt`).	Must	`profile.mfa.manage`	REQ-PROF-26, BR-PROF-11
FR-PROF-35	The system shall keep the authenticator key only in encrypted form and backup codes only in hashed form, and never show either again after setup.	Must	`profile.mfa.manage`	REQ-PROF-27, BR-PROF-11, BR-28
FR-PROF-36	The system shall, while setup is pending, block every screen and function except My Profile, Security Setup and these two-step actions, showing "Set up two-step verification to continue."	Must	`profile.mfa.manage`	REQ-PROF-28, BR-03
FR-PROF-37	The system shall show "Two-step on" on My Profile once two-step verification is on, and "Two-step off" otherwise.	Must	`profile.view`	REQ-PROF-02, REQ-PROF-25

ID	Requirement	Priority	Permission	BRD trace
FR-PROF-38	The system shall let a user with two-step verification on replace all backup codes with 10 new ones, shown once, after the current password (5 attempts in 5 minutes) and a valid code, making every old code stop working; refused with "Authenticator MFA is not on." when two-step is off.	Could	`profile.mfa.manage`	REQ-PROF-29
FR-PROF-39	The system shall let a user with two-step verification on check a code without using it up; wrong codes count toward the lock of FR-PROF-25.	Could	`profile.mfa.manage`	REQ-PROF-29 (BRD SCREEN 3 note)

> ⚠️ To be confirmed: FR-PROF-38 and FR-PROF-39 exist as server actions only; no screen offers them yet (BRD SCREEN 3 note).

Use case: UC-PROF-04 – Turn on two-step verification

Item	Detail
Actor	Signed-in user with setup pending
Preconditions	Signed in; two-step verification off; `profile.mfa.manage` (Always On)
Trigger	System opens Security Setup after sign-in
Main flow	1. User enters the current password and clicks **Show my QR code**. 2. System checks the password. 3. System shows the key and QR code (two-step still off). 4. User scans the code and enters the first 6-digit code, clicks **Verify and turn on**. 5. System checks the code and turns two-step on. 6. System shows 10 backup codes once. 7. My Profile shows "Two-step on".
Alternate flows	A1: QR code cannot be scanned → user copies the key with **Copy key**. A2: user downloads `taskflow-backup-codes.txt`.
Exceptions	E1: wrong password → "Current password is incorrect.". E2: 5 password attempts in 5 minutes → "Too many attempts. Try again in a few minutes.". E3: already on → "Authenticator MFA is already on. Turn it off first to set it up again.". E4: code before setup → "Start the setup first to get a key.". E5: wrong code → "The code is incorrect or has expired."; after 5 → "Too many wrong codes. Try again in a few minutes."
Postconditions	Two-step verification on; encrypted key and 10 hashed backup codes stored; all screens unblocked



Use case: UC-PROF-05 – Regenerate backup codes (server action, no screen)

Item	Detail
Actor	Signed-in user with two-step verification on
Preconditions	Two-step verification on; `profile.mfa.manage` (Always On)

Item	Detail
Trigger	⚠️ To be confirmed: no screen offers this action yet
Main flow	1. User supplies the current password and a valid code. 2. System checks the password (5 attempts in 5 minutes) and that two-step is on. 3. System checks the code. 4. System replaces all backup codes with 10 new ones, shown once.
Alternate flows	–
Exceptions	E1: two-step off → "Authenticator MFA is not on.". E2: wrong password → "Current password is incorrect.". E3: wrong code → "The code is incorrect or has expired."
Postconditions	Every old backup code stops working

Field validations

Field	Rule	Error message
Current password	Must match the account password; 5 attempts in 5 minutes per account	"Current password is incorrect." / "Too many attempts. Try again in a few minutes."
6-digit code	Valid code of the new key; setup must have been started	"The code is incorrect or has expired." / "Start the setup first to get a key."
Code (all checks)	5 wrong codes in a row → 5-minute lock, shared with sign-in	"Too many wrong codes. Try again in a few minutes."

Business rules applied: BR-PROF-01, BR-PROF-05, BR-PROF-09, BR-PROF-10, BR-PROF-11; overall BR-03, BR-28.

Notifications / side effects

Event	Who is notified	Channel
Two-step verification turned on	Nobody; the user's screens are unblocked	–
Backup codes regenerated	Nobody	–

Messages

Situation	Message
Already on	"Authenticator MFA is already on. Turn it off first to set it up again."
Two-step not on (backup codes)	"Authenticator MFA is not on."
Blocked screen while setup pending	"Set up two-step verification to continue."
No permission	*Not applicable – Always On.*

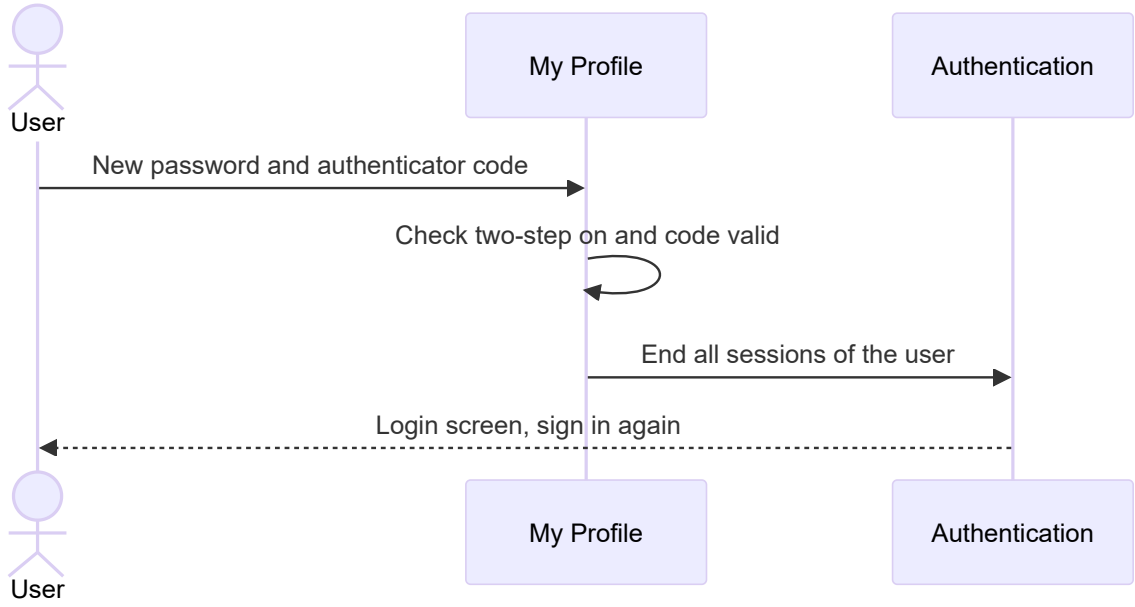
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5. Cross-Module Workflows

5.1 Password change and sign-in again (PROF → AUTH)

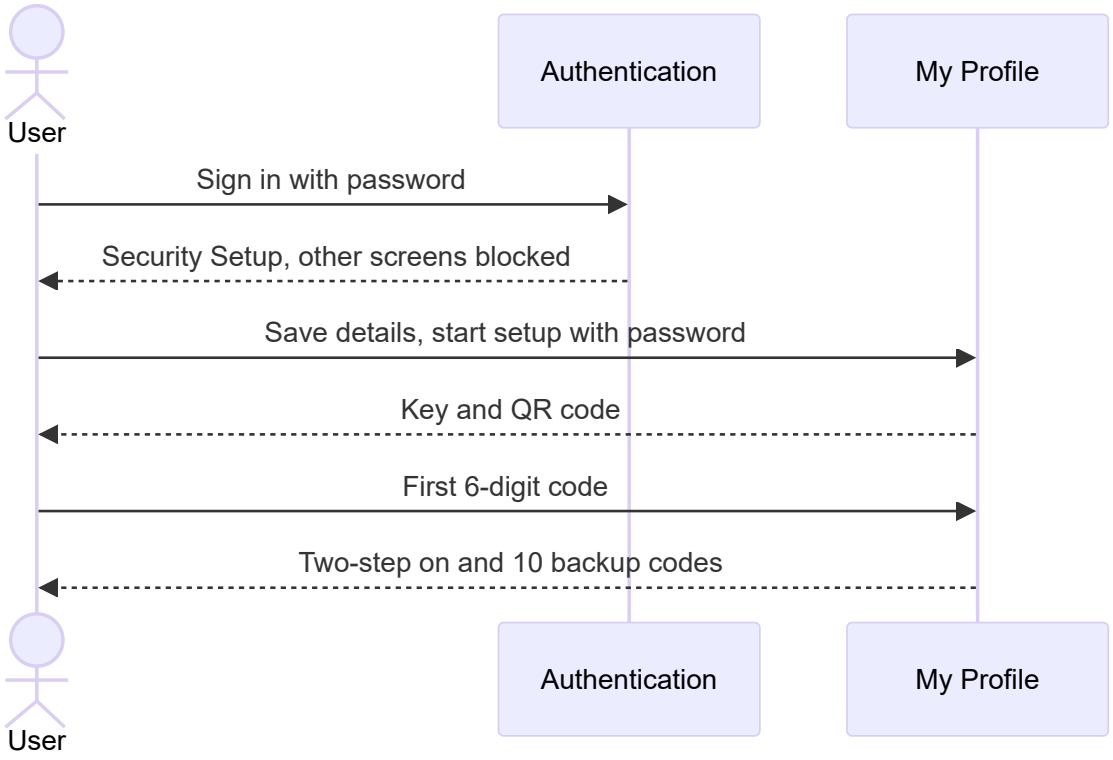
Step	Module	Actor / System	Action	FR reference
1	PROF	User	Enters new password twice and a code	FR-PROF-19, FR-PROF-20, FR-PROF-22
2	PROF	System	Checks two-step is on and the code; saves the password	FR-PROF-23, FR-PROF-24, FR-PROF-25

Step	Module	Actor / System	Action	FR reference
3	AUTH	System	Ends every session of the user on every device	FR-PROF-26
4	AUTH	User	Signs in again with the new password and a code	Overall FRD (`FR-AUTH-nn`)



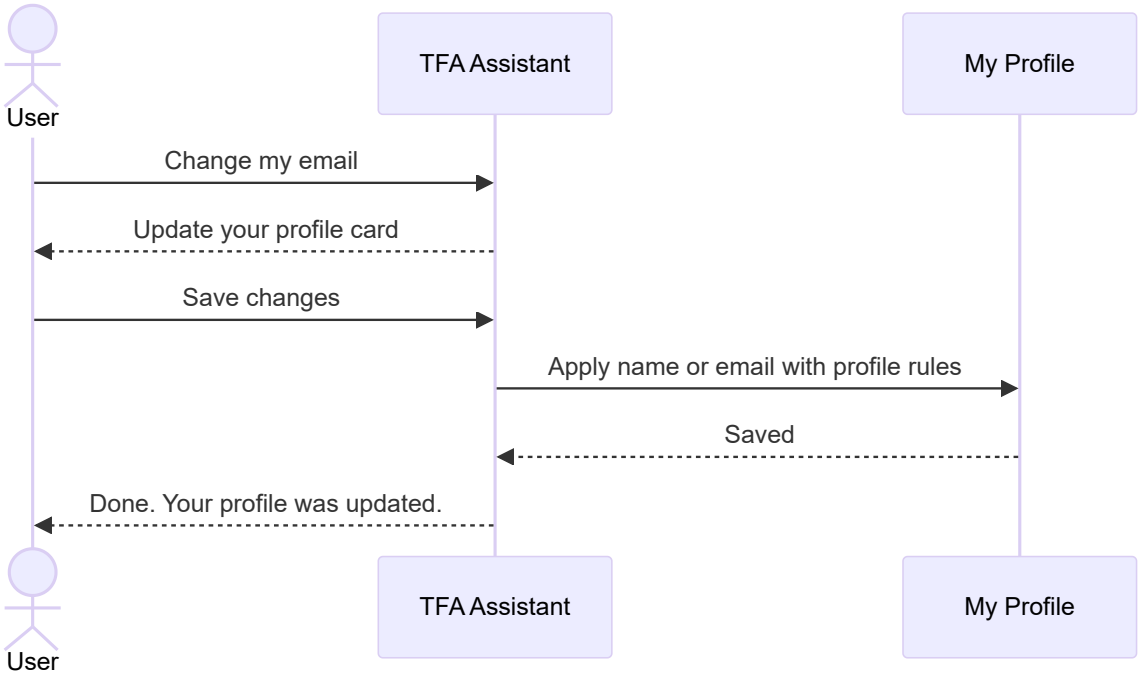
5.2 First sign-in and security setup (AUTH → PROF)

Step	Module	Actor / System	Action	FR reference
1	AUTH	User	Signs in without two-step verification	Overall FRD (`FR-AUTH-nn`)
2	PROF	System	Blocks all screens except My Profile and Security Setup	FR-PROF-36
3	PROF	User	Confirms name, email and mobile (Security Setup step 1)	FR-PROF-15
4	PROF	User	Starts setup, enters first code, receives backup codes	FR-PROF-30 – FR-PROF-35
5	PROF	System	Shows "Two-step on" and unblocks the screens	FR-PROF-37



5.3 Own profile change through TFA (TFA → PROF)

Step	Module	Actor / System	Action	FR reference
1	TFA	User	Asks TFA to change own name or email	FRD-Assistant (`FR-TFA-nn`)
2	TFA	System	Checks the values with the profile rules and shows "Update your profile"	FR-PROF-16
3	TFA	User	Clicks **Save changes** on the card	FRD-Assistant
4	PROF	System	Saves name/email with the profile rules, mobile unchanged	FR-PROF-16



6. Reports & Exports

Not applicable for this sub-module.

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7. Notifications & Messages Catalogue

My Profile sends no in-app notification and no mail (BR-PROF-12). The main user-facing messages are:

Code	Trigger	Recipients	Channel	Text
M-PROF-01	Profile saved	Signed-in user	Toast	"Profile updated"
M-PROF-02	Profile load failed	Signed-in user	On screen	"Could not load your profile."
M-PROF-03	Concurrent profile change	Signed-in user	On screen	"Someone else changed this record while you were saving. Reload it and try again."
M-PROF-04	Password changed	Signed-in user	Login screen	"Your username or password was changed. Sign in again with your new details."
M-PROF-05	Password change without two-step	Signed-in user	On screen	"Set up two-step verification before changing your password."
M-PROF-06	5 wrong codes	Signed-in user	Popup / screen	"Too many wrong codes. Try again in a few minutes."
M-PROF-07	Screen opened while setup pending	Signed-in user	On screen	"Set up two-step verification to continue."
M-PROF-08	Setup started while two-step on	Signed-in user	On screen	"Authenticator MFA is already on. Turn it off first to set it up again."

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8. Data Requirements (functional view)

Data item	Description	Created by	Rules
Full name	Own display name	User	Required; trimmed; max 200 characters
Email	Own address on `@taskflow.com`	Administrator (created), User (edited)	Lower case; max 256; unique by part before `@`; deleted accounts' addresses reserved
Mobile number	Own contact number	User	Optional; `+` and 8–15 digits, no leading zero; may be shared, not with the same email
Username, employee code, role, projects, departments, managers, status	Administrator-managed account facts	Administrator (Employee Management)	Read-only on My Profile

Data item	Description	Created by	Rules
Password	Sign-in secret	User	6–128 characters; stored only as a salted hash; never shown
Session marker	Ends all sessions	System	Renewed on password change
Authenticator key	Second-factor secret	System at setup start	Encrypted; shown once
Backup codes	Recovery second factor	System when two-step turns on	10 codes, hashed, single use; all replaced on regeneration
Code failure counter	Brute-force protection	System	5 wrong codes → 5-minute lock; shared with sign-in

9. Non-Functional Requirements (functional impact)

ID	Requirement	Affects
NFR-PROF-01	Passwords stored only as salted hashes; never shown or logged	Change Password (FR-PROF-26)
NFR-PROF-02	Authenticator keys encrypted, backup codes hashed	Two-step actions (FR-PROF-35)
NFR-PROF-03	5 wrong codes → 5-minute lock; 5 password tries per 5 minutes	FR-PROF-25, FR-PROF-30, FR-PROF-38
NFR-PROF-04	Field errors shown while typing, next to the field	FR-PROF-06, FR-PROF-20, FR-PROF-27
NFR-PROF-05	Concurrent edits never silently overwrite each other	FR-PROF-13
NFR-PROF-06	My Profile opens in under 2 seconds for 95% of loads on a normal office network	UC-PROF-01

10. Assumptions & Open Issues

Type	Description	Owner
Assumption	Every user has a smartphone with a standard time-based authenticator app	Jojo George
Assumption	All staff email addresses use the single domain `@taskflow.com`	Jojo George
Assumption	A user cannot turn two-step verification off; the route back is an administrator reset in Employee Management	Jojo George
Open issue	⚠️ To be confirmed: FR-PROF-38 (regenerate backup codes) and FR-PROF-39 (check a code) have no screen yet	Jojo George

Type	Description	Owner
Open issue	⚠️ To be confirmed: TFA password guidance and the "Change your password" help topic still mention the current password and say only other sessions end; they should match FR-PROF-22 and FR-PROF-26 (BRD R-PROF-05)	Jojo George
Open issue	⚠️ To be confirmed: the employee edit in Employee Management accepts a new password with only `employees.edit`, outside the code-protected flow of FR-PROF-22; decide whether this route should remain	Jojo George
Open issue	⚠️ To be confirmed: the global `BR-nn` numbers cited here follow the overall BRD v1.0 numbering	Jojo George

11. Requirements Traceability Matrix

Test cases are the automated test suites (backend test classes and frontend `*.spec.ts` files) that cover the behaviour; the mapping was checked against the test sources.

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-PROF-01	FR-PROF-01	My Profile	UC-PROF-01	`UserProfileEndpointsTests`
REQ-PROF-02	FR-PROF-02, FR-PROF-03, FR-PROF-37	My Profile	UC-PROF-01	`UserProfileEndpointsTests`, `profile.component.spec.ts`, `profile.component.layout.spec.ts`
REQ-PROF-03	FR-PROF-04	My Profile	UC-PROF-01	`UserProfileEndpointsTests`, `profile.component.spec.ts`
REQ-PROF-04	FR-PROF-05	My Profile	UC-PROF-02	`UpdateOwnProfileRequestValidatorTests`, `UserProfileEndpointsTests`
REQ-PROF-05	FR-PROF-06, FR-PROF-07, FR-PROF-13	My Profile	UC-PROF-02	`UpdateOwnProfileRequestValidatorTests`, `UserProfileEndpointsTests`, `profile.component.spec.ts`, `profile.component.layout.spec.ts`
REQ-PROF-06	FR-PROF-08	My Profile	UC-PROF-02	`EmailDomainTests`, `profile.component.spec.ts`
REQ-PROF-07	FR-PROF-09	My Profile	UC-PROF-02	`UserProfileEndpointsTests`, `EmailDomainTests`
REQ-PROF-08	FR-PROF-10	My Profile	UC-PROF-02	`ProfileMobileNumberEndpointsTests`
REQ-PROF-09	FR-PROF-11	My Profile	UC-PROF-02	`ProfileMobileNumberEndpointsTests`, `profile.component.layout.spec.ts`

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-PROF-10	FR-PROF-12	My Profile	UC-PROF-02	`profile.component.spec.ts`
REQ-PROF-11	FR-PROF-14	My Profile	UC-PROF-01	`profile.component.spec.ts`
REQ-PROF-12	FR-PROF-15	My Profile, Security Setup	UC-PROF-01, UC-PROF-02	`MfaSetupGateEndpointsTests`
REQ-PROF-13	FR-PROF-16	My Profile (via TFA card)	UC-PROF-02	`AssistantActionServiceTests`
REQ-PROF-14	FR-PROF-19	Change Password	UC-PROF-03	`ChangePasswordRequestValidatorTests`, `UserProfileEndpointsTests`
REQ-PROF-15	FR-PROF-18, FR-PROF-20, FR-PROF-28	Change Password	UC-PROF-03	`profile.component.layout.spec.ts`, `change-password.component.spec.ts`
REQ-PROF-16	FR-PROF-21	Change Password	UC-PROF-03	`change-password.component.spec.ts`
REQ-PROF-17	FR-PROF-22	Change Password	UC-PROF-03	`MfaStepUpEndpointsTests`, `change-password.component.spec.ts`
REQ-PROF-18	FR-PROF-23	Change Password	UC-PROF-03	`UserProfileEndpointsTests`
REQ-PROF-19	FR-PROF-24	Change Password	UC-PROF-03	`MfaServiceTests`, `MfaStepUpEndpointsTests`
REQ-PROF-20	FR-PROF-25	Change Password	UC-PROF-03	`MfaServiceTests`, `MfaSignInEndpointsTests`
REQ-PROF-21	FR-PROF-26	Change Password	UC-PROF-03	`MfaStepUpEndpointsTests`, `change-password.component.spec.ts`
REQ-PROF-22	FR-PROF-27, FR-PROF-28	Change Password	UC-PROF-03	`change-password.component.spec.ts`
REQ-PROF-23	FR-PROF-30, FR-PROF-31	Security Setup	UC-PROF-04	`MfaEnrolmentEndpointsTests`
REQ-PROF-24	FR-PROF-32	Security Setup	UC-PROF-04	`MfaEnrolmentEndpointsTests`

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-PROF-25	FR-PROF-33, FR-PROF-37	Security Setup	UC-PROF-04	`MfaEnrolmentEndpointsTests`
REQ-PROF-26	FR-PROF-34	Security Setup	UC-PROF-04	`MfaEnrolmentEndpointsTests`
REQ-PROF-27	FR-PROF-35	Security Setup	UC-PROF-04	`MfaServiceTests`, `TotpTests`
REQ-PROF-28	FR-PROF-36	All signed-in screens	UC-PROF-04	`MfaSetupGateEndpointsTests`
REQ-PROF-29	FR-PROF-38, FR-PROF-39	– (server only)	UC-PROF-05	`MfaServiceTests`, `MfaEnrolmentEndpointsTests`, `MfaStepUpEndpointsTests`
BR-PROF-01	FR-PROF-01	My Profile, Change Password, Security Setup	UC-PROF-01 – UC-PROF-05	`UserProfileEndpointsTests`
BR-PROF-02	FR-PROF-03, FR-PROF-05	My Profile	UC-PROF-02	`UpdateOwnProfileRequestValidatorTests`, `UserProfileEndpointsTests`
BR-PROF-03	FR-PROF-08, FR-PROF-09	My Profile	UC-PROF-02	`UserProfileEndpointsTests`, `EmailDomainTests`
BR-PROF-04	FR-PROF-10	My Profile	UC-PROF-02	`ProfileMobileNumberEndpointsTests`
BR-PROF-05	§3.2, FR-PROF-15, FR-PROF-36	All	UC-PROF-01	`PermissionCatalogTests` (Always On entries), `MfaSetupGateEndpointsTests`
BR-PROF-06	FR-PROF-22	Change Password	UC-PROF-03	`MfaStepUpEndpointsTests`
BR-PROF-07	FR-PROF-23	Change Password	UC-PROF-03	`UserProfileEndpointsTests`
BR-PROF-08	FR-PROF-26	Change Password	UC-PROF-03	`MfaStepUpEndpointsTests`
BR-PROF-09	FR-PROF-25, FR-PROF-39	Change Password, Security Setup	UC-PROF-03, UC-PROF-04	`MfaServiceTests`, `MfaStepUpEndpointsTests`
BR-PROF-10	FR-PROF-24	Change Password, Security Setup	UC-PROF-03	`MfaServiceTests`

BRD item	FRD requirement(s)	Screen	Use case	Test case
BR-PROF-11	FR-PROF-34, FR-PROF-35	Security Setup	UC-PROF-04	`MfaEnrolmentEndpointsTests`
BR-PROF-12	FR-PROF-17, FR-PROF-29	My Profile, Change Password	UC-PROF-02, UC-PROF-03	No automated test found
BR-PROF-13	FR-PROF-16	My Profile (via TFA card)	UC-PROF-02	`AssistantActionToolsTests`

Untraced BRD items: 0 (29 of 29 `REQ-PROF-nn`, 13 of 13 `BR-PROF-nn`).

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12. Approvals

Role	Name	Decision	Date
Product owner	Jojo George	Pending	–
Reviewer	Jojo George	Pending	–

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